

Ratchet: Literature-Only Transformer Optimization Survey

Ratchet project

2026-08-29

1 Scope

Ratchet preserves an authoritative full-transformer evaluator and prepares a future Intel-XPU research workflow. The present workspace has no qualified PyTorch XPU runtime. This paper therefore separates reviewed literature from project evidence and makes no project result claim.

2 Reviewed literature

This paper reports literature claims only, not measurements from this workspace. The reviewed primary-source set is [1, 3, 2, 5, 4, 6, 7, 8, 9]. The sources motivate future, hardware-gated investigation of IO-aware attention, compiler baselines, target-specific autotuning, and Intel XPU programming paths. They do not establish performance for this project or its unavailable runtime.

3 Catalogue summary

Catalogue fact	Value
Immutable experiment events	0
Reviewed primary sources	9

4 Evidence status

The immutable catalogue contains zero experiment events. ENV-0001 records an unavailable PyTorch XPU runtime, so this build generated no candidate, performed no compilation, and collected no correctness, timing, memory, trace, or speedup result. Consequently this document is a reproducible literature survey scaffold, not an empirical optimization report.

9 reviewed primary sources
Verified immutable archive projection
0 experiment events
Literature synthesis only; no empirical conclusions

Figure 1: Evidence boundary for this literature-only build.

References

- [1] Jason Ansel, Edward Yang, Horace He, N. Gimeshein, Animesh Jain, et al. Pytorch 2: Faster machine learning through dynamic python bytecode transformation and graph compilation. In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2*, 2024.
- [2] Tri Dao. Flashattention-2: Faster attention with better parallelism and work partitioning. In *International Conference on Learning Representations*, 2024.
- [3] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. Flashattention: Fast and memory-efficient exact attention with IO-awareness. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- [4] Intel Corporation. Programming intel xmx using sycl joint matrix extension, 2024.
- [5] Intel Corporation. Intel extension for pytorch, 2026. Archived repository read 2026-08-29; repository archived 2026-03-30; retirement notice says active development and quarterly releases had already ceased.
- [6] Intel Corporation. Intel xpu backend for triton, 2026. Read 2026-08-29; architecture: <https://github.com/intel/intel-xpu-backend-for-triton/blob/main/docs/ARCHITECTURE.md>.
- [7] PyTorch Contributors. torch.xpu and getting started on intel gpu, 2026. Read 2026-08-29; companion installation guide: https://docs.pytorch.org/docs/main/notes/get_start_xpu.html.
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- [9] Marcin Spoczynski, Daniel Fleischer, Moshe Berchansky, Gabriela Ben-Melech Stan, Shira Guskin, Weilin Xu, Adam Siemieniuk, and Alexander Heinecke. Xe-forge: Multi-stage LLM-powered kernel optimization for intel GPU. *arXiv preprint arXiv:2605.26118*, 2026.